Improving Flight Cancellation Insurance Profitability Through Weather-Informed Pricing

1. The Business Problem

Imagine an insurance company offering flight cancellation insurance at a fixed premium of 6.5% of the ticket price. Although this pricing strategy is simple, it does not account for differences in cancellation risk between flights. Some flights have a very low probability of cancellation, while others face considerably higher risks, particularly when adverse weather conditions are expected.

This creates a profitability problem for the insurer. Premiums collected from low-risk flights may comfortably cover potential claims, whereas high-risk flights can generate substantial losses. The insurer is effectively charging the same price for policies that carry very different levels of risk.

The problem may be particularly relevant for customers purchasing insurance shortly before departure. These customers have access to more recent weather forecasts and may be more inclined to purchase insurance when adverse weather conditions are expected. If this results in a higher concentration of high-risk customers among late purchases, the insurer could face additional claims without receiving higher premiums.

2. The Proposed Solution: Weather-Informed Pricing

A potential solution is to move away from fixed-premium pricing towards a risk-based pricing strategy that incorporates weather information.

By using weather information to improve cancellation risk predictions, the insurer can identify flights with a higher likelihood of cancellation. Premiums can then be adjusted to reflect the estimated risk. In particular, customers purchasing insurance closer to departure could be charged higher premiums when more recent weather information indicates an elevated cancellation risk.

To examine the potential benefits of this approach, two pricing strategies are compared:

- **Booking-time pricing:** Uses information available when the flight is booked to estimate cancellation risk.
- **Live-weather pricing:** Incorporates actual departure-day weather information to produce more informed cancellation risk estimates.

The aim is to determine whether more accurate risk differentiation allows the insurer to charge premiums that better reflect the underlying risk and consequently improve profitability.

3. Identifying High-Risk Flights

The first step is to examine how accurately the two models identify flights with an increased cancellation risk. Flights are divided into 100 risk groups, ranging from the lowest predicted cancellation risk (group 1) to the highest (group 100).

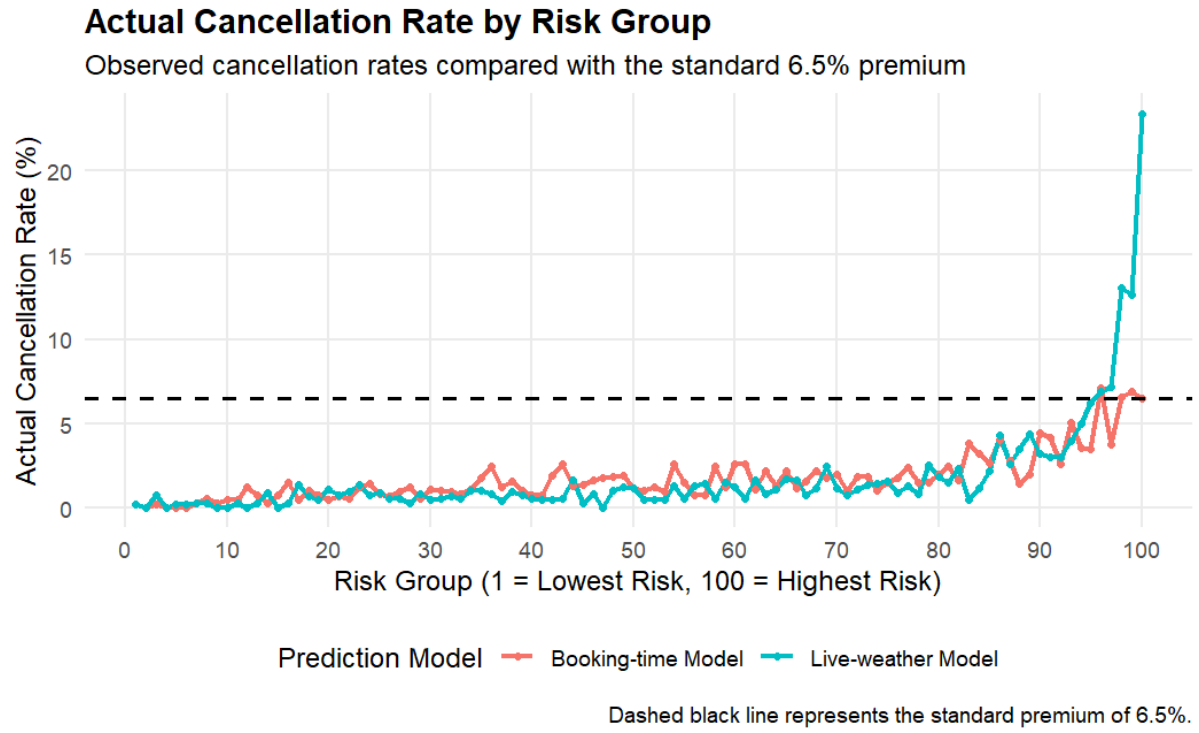


Figure 1. Actual cancellation rates by risk group for the booking-time and live-weather models. The dashed line represents the standard premium of 6.5%.

The results show that both models identify an increasing cancellation risk across the higher risk groups. However, the live-weather model distinguishes the highest-risk flights more clearly. In particular, the actual cancellation rate rises sharply in the highest risk groups, reaching considerably higher levels than under the booking-time model.

This demonstrates the potential value of incorporating more recent weather information. The live-weather model provides a stronger distinction between relatively low-risk and high-risk flights, allowing the insurer to identify groups that may require higher premiums.

4. Adjusting Premiums to Reflect Risk

Once higher-risk flights have been identified, the insurer can use the estimated cancellation rates to determine more appropriate premiums for each risk group.

Rather than charging every customer the standard premium of 6.5%, the pricing strategy assigns premiums based on the cancellation rates observed in the calibration sample. Customers in low-risk groups continue to pay the standard premium, while customers in higher-risk groups may be charged more.

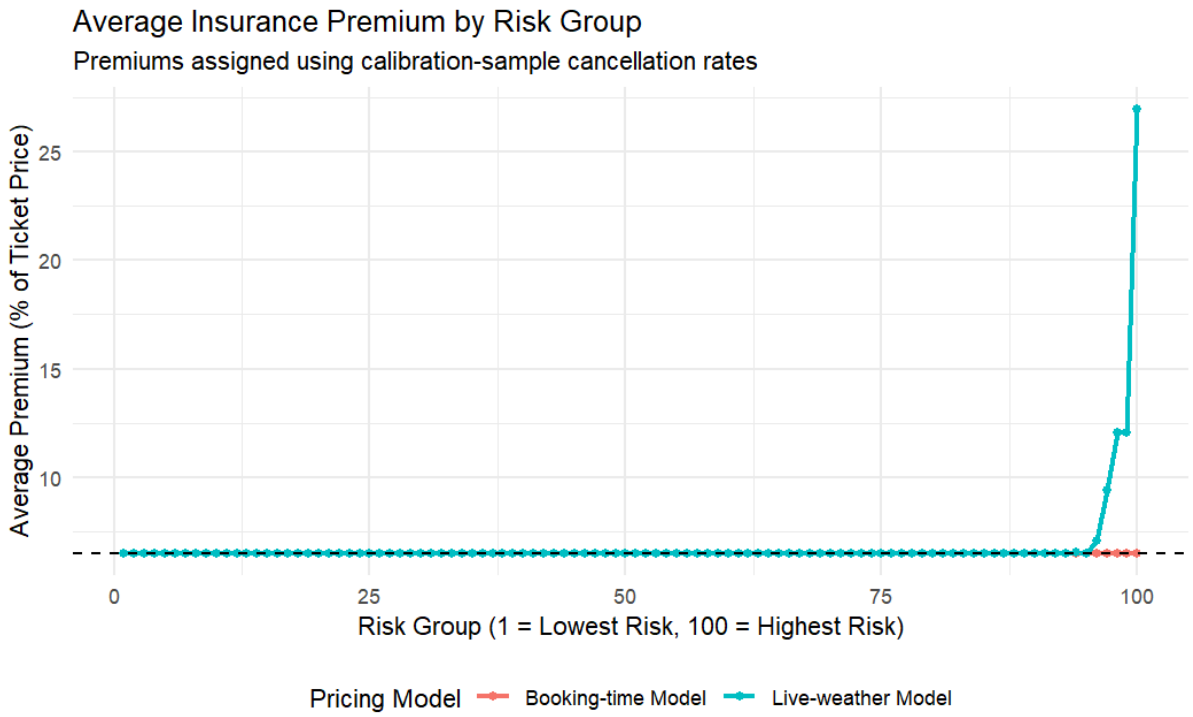


Figure 2. Average insurance premiums assigned to each risk group using the calibration-sample cancellation rates.

The results show that the booking-time model assigns premiums close to the standard 6.5% rate across the risk groups. In contrast, the live-weather model identifies substantially higher cancellation risks in the highest groups and consequently assigns higher premiums to these customers.

This illustrates how more detailed weather information can translate into differentiated pricing. Instead of charging a uniform premium, the insurer can increase prices for customers whose flights carry a substantially higher estimated cancellation risk.

5. The Potential Impact on Insurer Profitability

The final step is to examine what this pricing strategy could mean for the insurer's overall profitability.

The simulation considers different proportions of customers purchasing insurance shortly before departure. Early purchases are assigned booking-time pricing, while late purchases are assigned live-weather pricing. The resulting profits are compared with the existing standard-pricing strategy.

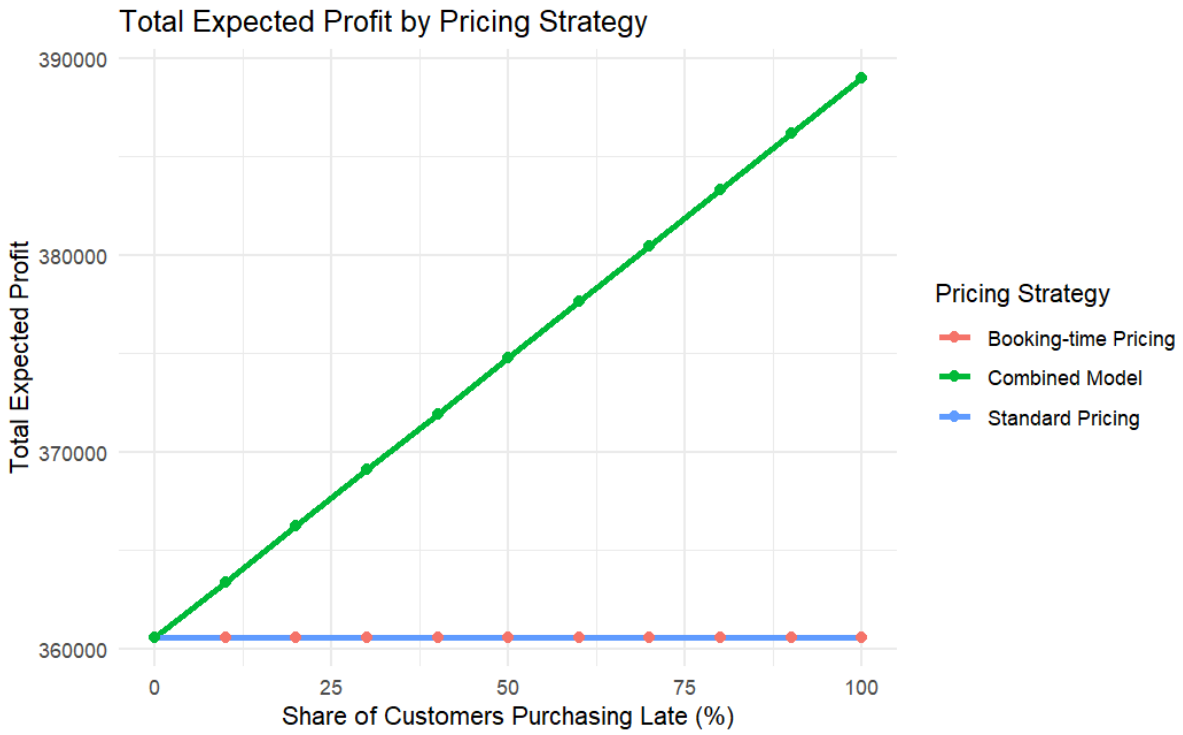


Figure 3. Simulated total insurer profit under different shares of late purchases. The combined model applies booking-time pricing to early purchases and live-weather pricing to late purchases.

The results indicate that the combined pricing strategy becomes increasingly profitable as the share of late purchases rises. This is because a larger proportion of customers is priced using the live-weather model, which generates higher average profit per policy in the evaluation sample.

Under the simulated assumptions, the combined strategy therefore offers a potential improvement over both the existing fixed-premium approach and a strategy that relies exclusively on booking-time information.

6. Conclusion and Limitations

The case illustrates how weather information could help an insurer improve the profitability of flight cancellation insurance. By identifying flights with a higher cancellation risk and adjusting premiums accordingly, the insurer can better align the price of insurance with the risk being covered.

The results suggest that live-weather information has the potential to improve risk differentiation, increase premiums for high-risk flights and raise overall simulated profitability, particularly when a larger share of customers purchases insurance closer to departure.

However, the results should be interpreted in light of several simplifying assumptions. First, the simulation assumes that all customers purchase insurance regardless of the premium charged. In reality, higher premiums may discourage some customers from purchasing insurance. Second, the late buyers in the simulation are treated as a random share of the customer population. They are not specifically selected from the highest-risk flights. If late buyers are disproportionately likely to insure flights with adverse weather forecasts, the actual profitability impact could differ. Finally, actual departure-day weather is used as a proxy for the more recent weather information that could be available to the insurer close to departure.

Despite these limitations, the simulation demonstrates the potential commercial value of incorporating weather information into insurance pricing. It provides a basis for a more sophisticated pricing strategy that accounts for differences in cancellation risk rather than relying on a single fixed premium.